

Lab Note: What We Have Learned About Links in HSF

Ben Bray
Independent Researcher
sjbbray@gmail.com

March 11, 2026

Purpose

This note records the current state of understanding of *links* in the Hilbert Substrate Framework (HSF). The goal is not to present a finished theory, but to state clearly what the framework and current evidence package actually support.

Main conclusion

The central lesson so far is:

A link in HSF is not merely adjacency or bookkeeping. A real link is an interface sector: a structured Hilbert-space register that carries irreducible relational structure between subsystems.

More sharply, the current work points toward the claim that a physically meaningful link must behave as a *bidirectional interface*, not merely as an arbitrary shared space with two algebras stuffed inside it.

1. A link is an interface, not just an edge

HSF III frames links as the interface degrees of freedom through which subsystems exchange influence. The narrow claim there is that if a link is to support independent left and right endpoint actions, then it cannot just be treated as a small undifferentiated carrier space. Instead, it must factor as

$$\mathcal{H}_{\text{link}} \cong V \otimes \bar{V},$$

with the left endpoint acting as $\rho(g) \otimes I$ and the right endpoint acting as $I \otimes \bar{\rho}(g)$. In that sense, a link is an *interface register* with independent endpoint structure, not merely an abstract graph edge.

2. Bidirectional links require a real size threshold

The strongest algebraic result so far is the threshold claim:

$$d_B \geq N^2$$

for an $SU(N)$ bidirectional link. In the minimal case,

$$d_B = N^2,$$

the link can host commuting left and right endpoint actions while still filling the full matrix algebra under the no-signaling plus no-forgetting proxies.

For $SU(3)$, this gives the minimal composite-link threshold

$$d_B = 9 = 3^2.$$

This matters because it says a genuine bidirectional link cannot, in general, be “small and simple” in the naive sense. A single qutrit link can support an endpoint algebra, but it cannot host two clean independent commuting endpoints. The minimal bidirectional object is already composite.

3. No-signaling alone is too weak

One of the most important negative lessons is that no-signaling, if implemented only as endpoint commutativity, does *not* force the right kind of link. It allows reducible or slack-heavy solutions with large commutants and many inert degrees of freedom.

So a link is not defined just by “left and right commute.” That condition is necessary as an interface proxy, but insufficient. Something else has to suppress refolding slack and underused dead directions.

4. No-refolding points toward committed interfaces

The current working definitions suggest that no-refolding is the pressure that turns a merely possible interface into a committed one. The key idea is:

Once structure exists, the tensor split associated with a working link is not freely mutable.

Operationally, the no-refolding witness distinguishes between coherent propagated internal rearrangements and incoherent refolding. The evidence script treats this as a concrete structural restriction, not just a verbal principle. The physical meaning is that if refolding were free, there would be no stable answer to the question:

“Which subsystem am I actually coupled to through this link?”

So no-refolding is what begins to make a link *committed* rather than merely representable.

5. Finite bandwidth suggests that links should be thin

Finite bandwidth has not yet been finalized as a full dynamical theorem, but the operational direction is clear. The intended picture is that a link cannot carry unlimited simultaneous influence. Instead, it should support only a limited number of strong operator channels at a time.

The HSF-native proxy is an influence map

$$\Phi_{A \rightarrow L},$$

constructed by poking one side and reading the induced response on the link endpoint. The singular values of this response map then measure how many independent influence directions are actually active.

This leads to the emerging interpretation:

The rank of a link is the effective number of active relational channels crossing the interface.

Low-rank links are thin, selective, and structured. High-rank links are broad, overstuffed, or scrambling.

6. Gauge structure strengthens the picture

The composite $SU(3)$ link result goes beyond pure algebra. The current evidence package shows that a composite edge with $d_B = 9$ can satisfy local gauge-compatibility conditions to numerical precision, and that the conjugate pairing $3 \otimes \bar{3}$ admits a nonempty Gauss-law singlet sector.

This matters because it suggests that the link is not just a carrier of transmission, but also the place where color information and gauge-compatible relational structure can live. In the minimal singlet witness, the color information is carried by the links rather than the sites themselves.

7. What the current evidence package supports

In limited witness form, the current evidence package supports the following statements about links:

1. Minimal bidirectional links require at least $d_B = N^2$ to carry clean independent left/right inherited structure.
2. A composite $SU(3)$ edge can satisfy exact local attachment/gauge compatibility.
3. No-refolding can be realized as a concrete structural restriction rather than only a verbal principle.
4. An explicit transmitting chain can carry nonzero signal while remaining insensitive to added uncoupled slack.
5. Adjacent inherited structures in such a chain can be locally compatible bond by bond.

These are not yet the full theory of links, but they are enough to say that links in HSF are already more than abstract graph edges.

8. What we should *not* claim yet

The current materials do *not* yet justify the stronger claims that:

- links have fully emerged from a bare unfactorized Hilbert substrate,
- finite bandwidth has been turned into a final dynamical law,
- large many-body systems automatically maintain thin committed link sectors,
- or the framework already has full confinement observables or a global gauge law over arbitrary transmitting chains.

So the correct status is: *evidence and mechanism*, not finished derivation.

9. Best current definition

The best compact definition I can give at this stage is:

A link in HSF is an emergent relational interface sector: a structured Hilbert-space register that mediates influence between subsystems through a limited set of active channels, and which becomes physically meaningful when its endpoint structure is both independent and committed.

This includes four ideas at once:

- a. links are *real relational structure*,
- b. links are *interface sectors*, not just adjacency,
- c. links are *limited* by bandwidth-like channel count,
- d. links become *committed* under no-refolding.

10. What still needs to be understood

The open problems are now clearer:

1. How exactly should finite bandwidth be implemented as a dynamical pressure?
2. Can thin committed links emerge automatically in larger many-body simulations, rather than only in hand-structured witnesses?
3. How do link sectors combine into larger lattice or network structure?
4. When do links remain thin, and when do they bloat into generic entanglement spread?
5. How do particles or bound objects live *on* or *through* these links?

Working bottom line

At the present stage, the safest statement is:

HSF has not yet finished the theory of links, but it has already moved beyond the idea that a link is merely an abstract edge. The current algebraic and computational witnesses support the view that links are structured, bidirectional, bandwidth-limited interface sectors whose stable physical meaning depends on no-refolding-type commitment.